

Pratishtha Abrol

Platform & Infrastructure · SRE/DevOps · Backend Engineer

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PROFESSIONAL SUMMARY

Distributed systems engineer with 2+ years of production experience in **Kubernetes-based distributed systems**, **backend API development**, and **SRE practices**. Shipped independent features at Oracle, including a **21-second latency reduction** and a **FIPS 140-2 compliant DNS service** adopted by 2 internal teams. Currently building Ferrum, a production-grade **distributed webhook platform** progressing from **CI/CD + Kubernetes** through **SRE + zero-trust security**, to **platform engineering at scale**.

SKILLS

Languages: Python, Go, Bash, SQL, JavaScript, Java, C/C++

Containers & Orchestration: Docker, Kubernetes (EKS, Minikube), Helm, HPA, PVC

Cloud: Oracle Cloud Infrastructure, DigitalOcean

CI/CD & GitOps: GitHub Actions, Jenkins, GHCR, GitOps pipelines, blue-green deployment, canary release

Observability: Prometheus, Grafana, incident management

Infrastructure as Code: Docker Compose

Backend: REST API design, event-driven architecture, Redis, PostgreSQL, MongoDB, JWT/RBAC

OS & Networking: Linux, DNS, TCP/IP, FIPS 140-2

Tools: Git, GitHub, GitLab, k6

EXPERIENCE

Software Developer I

Oracle India

Jul 2024 – Apr 2026 (*Team Restructuring*)

Hyderabad (Remote | Hybrid)

- Owned **cluster controller logic** for a **distributed Kubernetes platform** (approx 20 pods, multiple replicas), managing workload orchestration and pod lifecycle across multi-node production environments. Designed and enforced **exactly one si-key assignment** across concurrent controller replicas, guaranteeing mutual exclusion and a single buffer pod at all times via **parallel processing** and synchronisation logic.
- Reduced **cluster scale-up latency by 21 seconds** (measured against production SLO baselines) by redesigning **scheduling algorithms** and **synchronization mechanisms**, hence directly improving downstream service reliability.
- Shipped a **FIPS 140-2 compliant DNS service** with zero-downtime cutover into the production **Kubernetes networking stack**, adopted by 2 internal teams, by implementing a **plugin chaining architecture** that satisfied compliance requirements without service interruption.
- Improved **observability** across platform by introducing structured log levels, reducing noise in production logs and contributed to **incident postmortems** and **reliability improvements** alongside cross-functional SRE and platform teams; participated in **on-call rotation**.

Platform Engineer

Ferrum Webhook Platform

May 2026

(*Personal Project*) Remote

- Achieved **135 req/s throughput at 0% error rate** (measured via **k6 load test**, 50 VUs over 60s) by architecting a **distributed webhook delivery system** {**Fast API** gateway + **async worker** + **PostgreSQL** + **Redis** queue} deployed on Kubernetes with **Horizontal Pod Autoscaling**, readiness/liveness probes, and resource limits.
- Instrumented full **Prometheus** observability layer across gateway and worker services, tracking request throughput, **p95 queue latency**, and event processing rate via **PromQL** queries.
- Built end-to-end **GitOps-aligned CI/CD pipeline** using **GitHub Actions**, automated image build, push to **GHCR**, and dynamic **Kubernetes** pull on deploy; full **Docker Compose** parity for local development.
- Reduced **webhook latency from 22ms to 1.2ms** (95% reduction) by implementing a **cache-aside pattern** on Redis with TTL-based expiry and explicit cache invalidation on every write
- Decoupled event ingestion from delivery by redesigning Ferrum into a **gateway + async worker architecture** over a **Redis queue**. Client response time dropped < 5ms regardless of downstream webhook endpoint latency, eliminating synchronous delivery failures under load.

Full-Stack Backend Developer

Contract, Automotive SaaS Client

Sep 2023 – Mar 2024

Remote

- Independently architected, deployed and delivered a **production MERN application** on **Digital Ocean App Platform**, multi-component deployment (backend, frontend, cron jobs) handling **200+ daily requests** across **20+ concurrent users** with **JWT authentication** and **role-based access control (RBAC)** and environment aware **CORS** configuration.
- Designed a **12-model MongoDB schema** with optimized aggregation pipelines, nested schemas (vehicle, customer, job, billing, and **indexed unique fields** supporting a multi-tier billing system with **CGST/SGST tax logic**, 4 payment modes, and **idempotent billing operations** preventing duplicate invoices.
- Automated end-to-end billing workflows {dynamic invoice generation, tax/discount logic, and **6 cron-driven batch scripts**} eliminating manual processing and reducing billing cycle time for a multi-tier SaaS product.
- Eliminated **race conditions** in high-concurrency job card numbering (measured: zero duplicate IDs under concurrent writes) by engineering a **timezone-aware, duplicate-safe sequential ID system** using IST offset logic.
- Built a **40+ component React frontend** with custom **route guards**, multi-step workflow state management, **PDF generation** (jsPDF + autotable), and 5+ report types.

DevOps and Analytics Intern

Oracle India

May 2023 – Jun 2023

Hyderabad (Remote)

- Built and deployed **Docker** images for microservices; orchestrated on **Kubernetes** and integrated with **Oracle Analytics Cloud (OAC)** for metrics visibility.
- Collaborated across cross-functional teams on service integration and deployment workflow standardization.

PROJECTS

Mini LinkedIn

MERN

- Full-stack job portal with dual-role RBAC (Applicant / Recruiter), dynamic application tracking, **Google OAuth**, resume upload, and **fuzzy search**; end-to-end **REST API design** and **MongoDB** schema ownership.
- Enforced data integrity via unique email/name validation across roles, cascade-deletion of applications on job removal, and persistent cross-role public profiles

Linux Terminal Emulator

C, Linux OS

- Built a **Unix shell from scratch** supporting command execution, **piping**, **I/O redirection**, interrupt handling, **process control**, and **IPC**
- Demonstrated systems programming depth at the OS level.

OPEN SOURCE

Outreachy Intern

TensorFlow

May 2021 – Aug 2021

Remote

- Contributed production **machine learning pipeline** components to **TensorFlow**'s infrastructure; followed **CI/CD** and code review workflows in a globally distributed open-source team. [\[source code\]](#)

PUBLICATION

Maximal Secret Reconstruction, Teleportation and Bell's Inequality

EPJD, 79, 11 (2025)

Center for Quantum Science and Technology, Center for Security, Theory, Algorithms and Security

[\[Article\]](#)

- Peer-reviewed journal paper exploring quantum information protocols, Bell inequalities, and secure state reconstruction. Demonstrates capacity for independent research, formal proof, and academic publishing — relevant for engineering orgs with high technical depth bars

EDUCATION

International Institute of Information Technology

Hyderabad, IN

B.Tech and Master of Science (by Research) in Computer Science

Aug. 2019 – Jul. 2025

- **Relevant Coursework:** Operating Systems, Distributed Systems, Computer Networks, Computer Architecture, Software Engineering & System Design, Algorithms & Data Structures, Database Management Systems, Quantum Information and Communication, Open Quantum Systems

CERTIFICATIONS

- Claude 101, Claude Code in Action Anthropic
- Open Source Software Development, Linux and Git Specialization Coursera